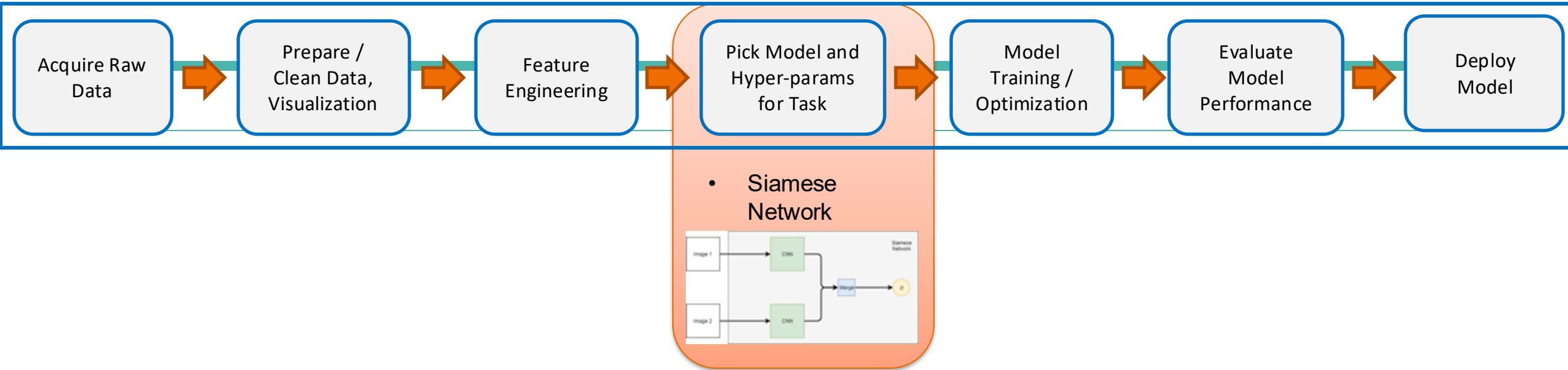


Learning Settings



Siamese and Loss Functions

Recognition Vs Verification

- Recognition: eg. K–Class recognition
 - Given a sample, decide which one of the K it is
- Verification: Given a sample and a class ID,
 - Say “YES” or “NO”
- What makes these two different?
 - Popular in many situations (Eg. Biometrics)
 - Face, Fingerprint
 - An Intra-class variation could be higher than inter-class
 - People change appearance over time.

Verification: Test/verification time

- Often done with nearest neighbors
 - Too much of computing?
- Find “K” nearest neighbors Find the distance to the samples from the class under question
 - Find Mean distance
 - Find Min Distance
- Variable Threshold: Subject specific
 - Some people have high variability in “signature”

Verification: Enrolment/Training

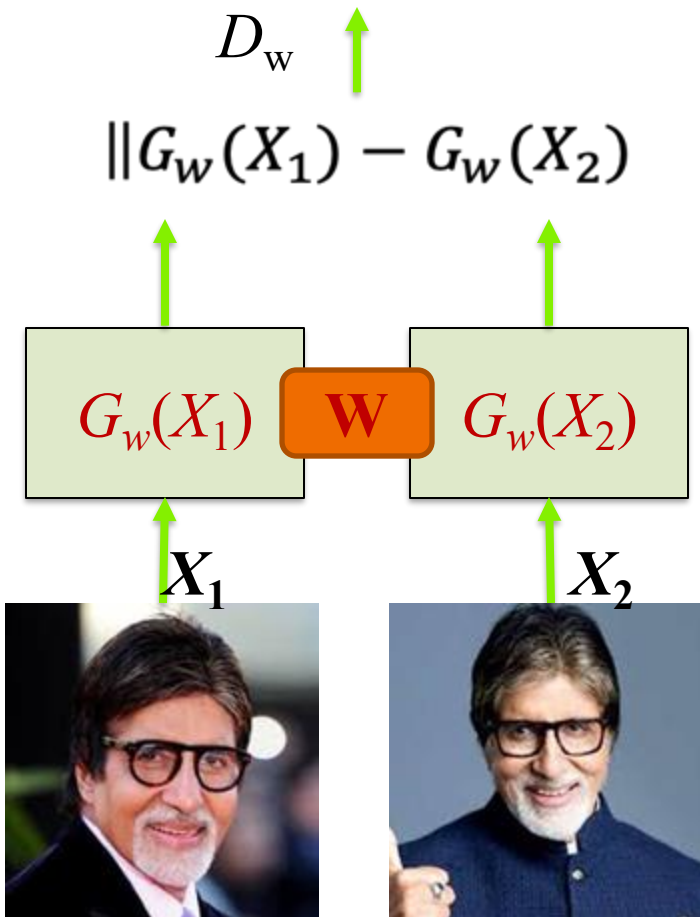
- Usually only small number of samples to add
 - One shot learning (single sample setting)
- Find a feature space that suits the local “geometry”
 - Metric learning
 - Aka finding an appropriate distance function

Points to Notice

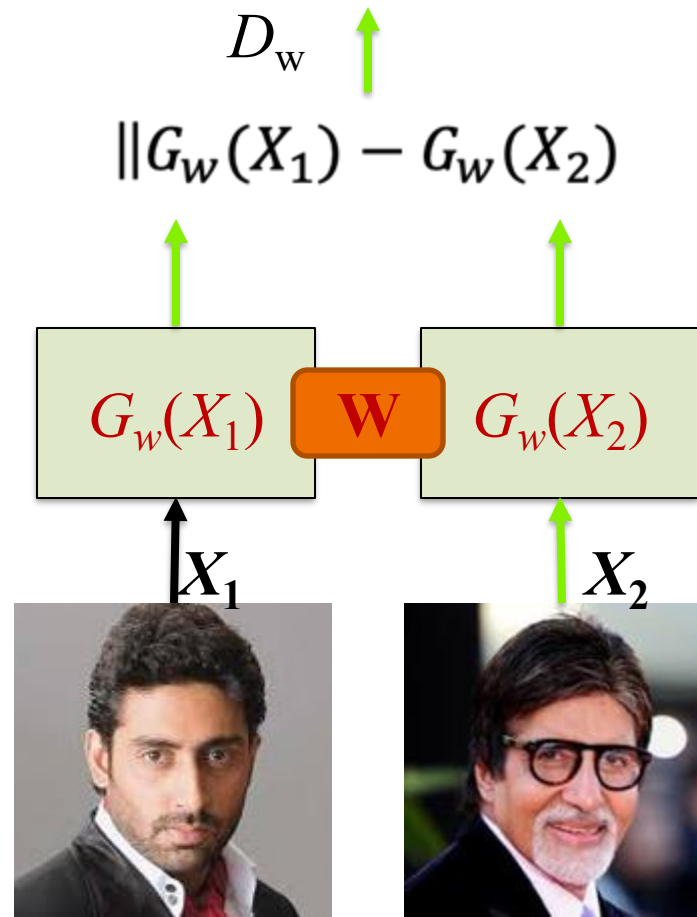
- Features that can capture “Fine Variation”
- Classification Vs Verification
- Learning with a “different style” of Supervision
- Contrastive Learning and Self Supervised Learning

Siamese Architecture/Loss

Make this smaller



Make this larger



- Only pair-wise Labels
- Similarity Metric:
 $D_w(X_1, X_2)$
- Have shared weights
- Training in batches

Historic

- **Input:** A pair of input signatures.
- **Output (Target):** A label, **0** for similar, **1** else.

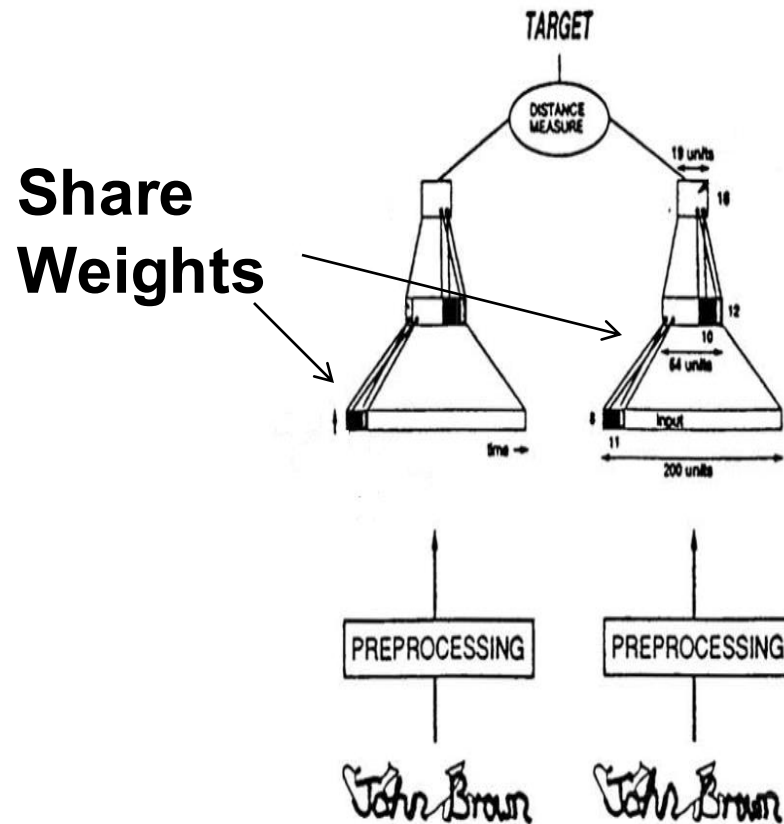
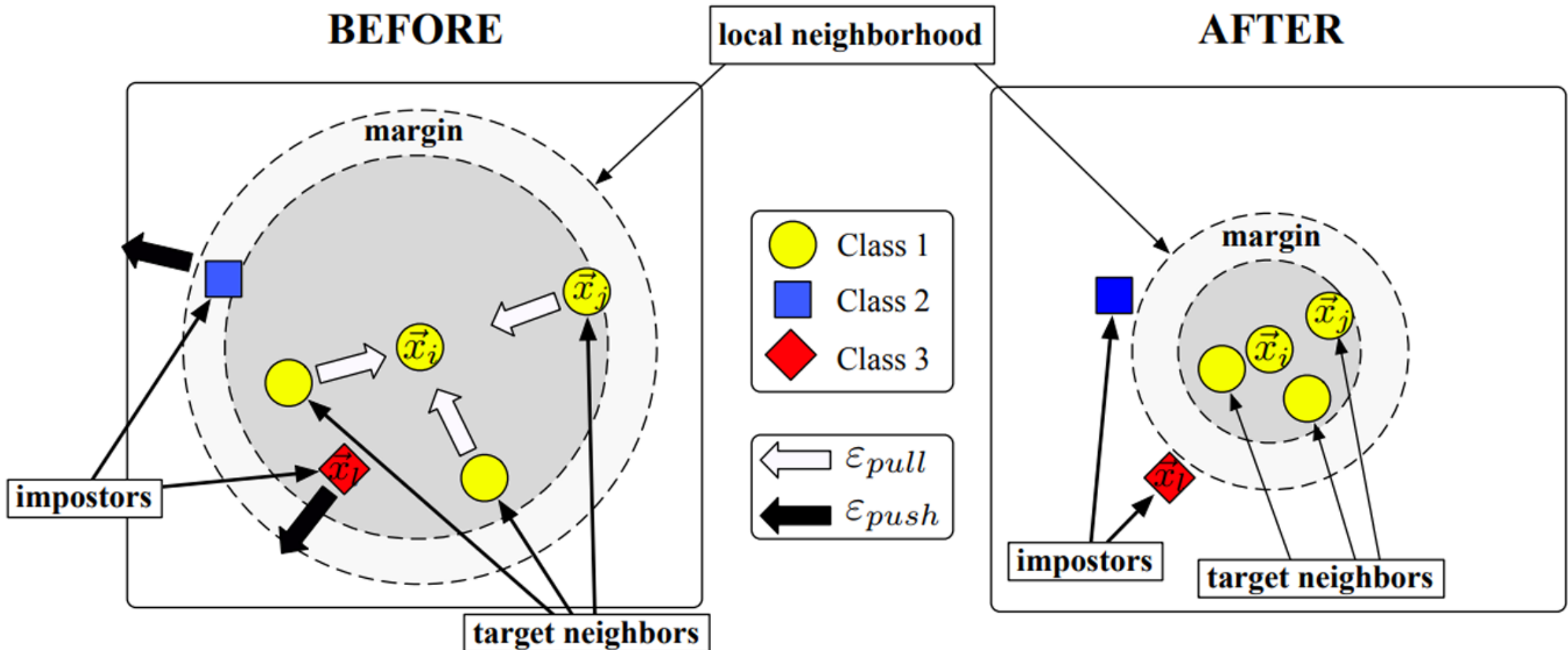


Image Source:
Google

Bromley, J., Bentz, J.W., Bottou, L., Guyon, I., LeCun, Y., Moore, C., Säckinger, E. and Shah, R., 1993. Signature Verification Using A "Siamese" Time Delay Neural Network. *IJPRAI*, 7(4), pp.669-688.

Historic: Large Margin Nearest Neighbour(LMNN)



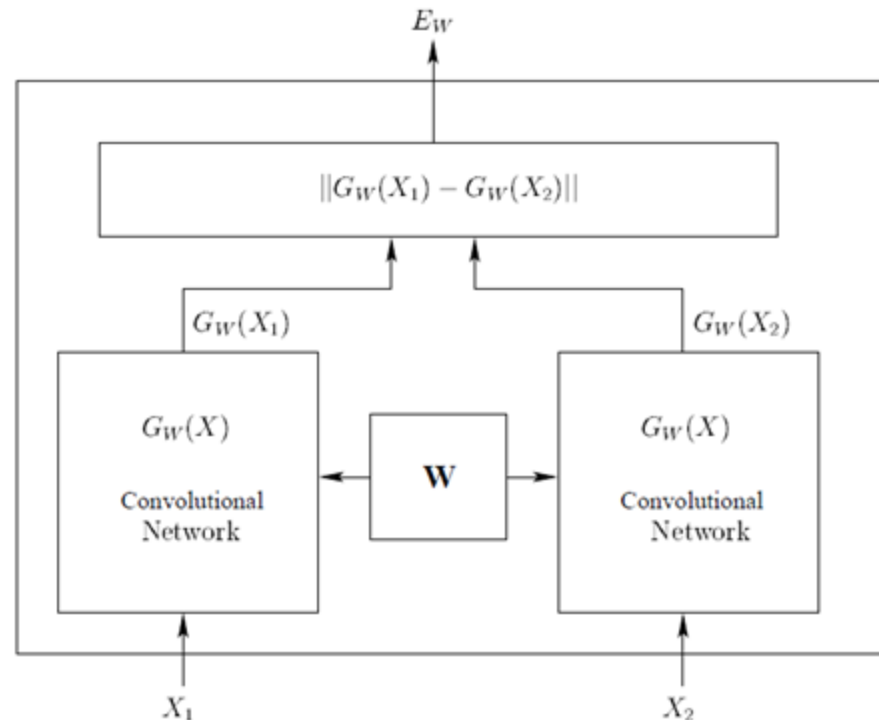
Intuition in Nearest Neighbor Setting

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Siamese Architecture

- Given a family of functions $G_W(X)$ parameterized by W , find W such that the similarity metric $D_W(X_1, X_2)$ is small for similar pairs and large for dissimilar pairs:-

$$D_W(X_1, X_2) = \|G_W(X_1) - G_W(X_2)\|$$

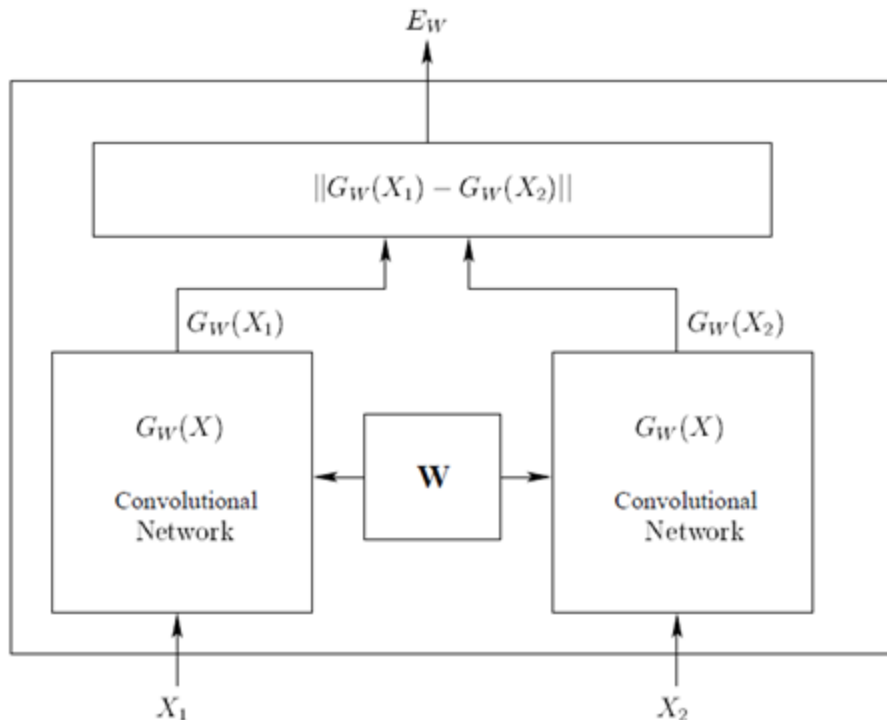


Siamese Architecture

- Given a family of functions $G_W(X)$ parameterized by W , find W such that the similarity metric $D_W(X_1, X_2)$ is small for similar pairs and large for dissimilar pairs:-

$$D_W(X_1, X_2) = \|G_W(X_1) - G_W(X_2)\|$$

$Y=0$ for “similar” and $Y=1$ for “dissimilar” pairs.



Loss function

$$\mathcal{L}(W) = \sum_{i=1}^p L(W, (Y, \vec{X_1}, \vec{X_2})^i)$$

Loss function for similar pairs

$$L(W, (Y, \vec{X_1}, \vec{X_2})^i) = (1 - Y)L_s(D_W^i) + YL_D(D_W^j)$$

Loss function for dissimilar pairs

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Contrastive Loss function

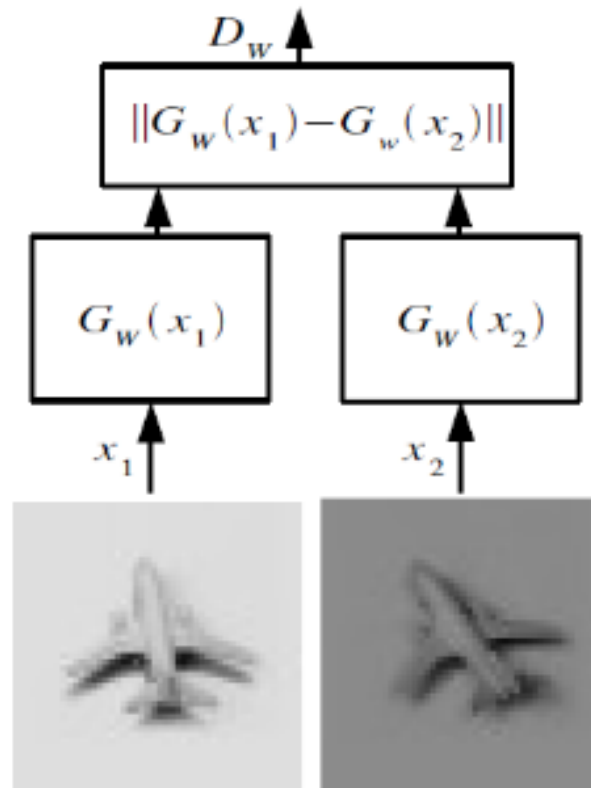
- Given $D_W(X_1, X_2)$ is the distance between the pair of samples, the contrastive loss function is defined as follows: ($Y=0$ for Similar and $Y=1$ for dissimilar pairs)

$$(1 - Y) \frac{1}{2} (D_W)^2 + (Y) \frac{1}{2} \{\max(0, m - D_W)\}^2$$

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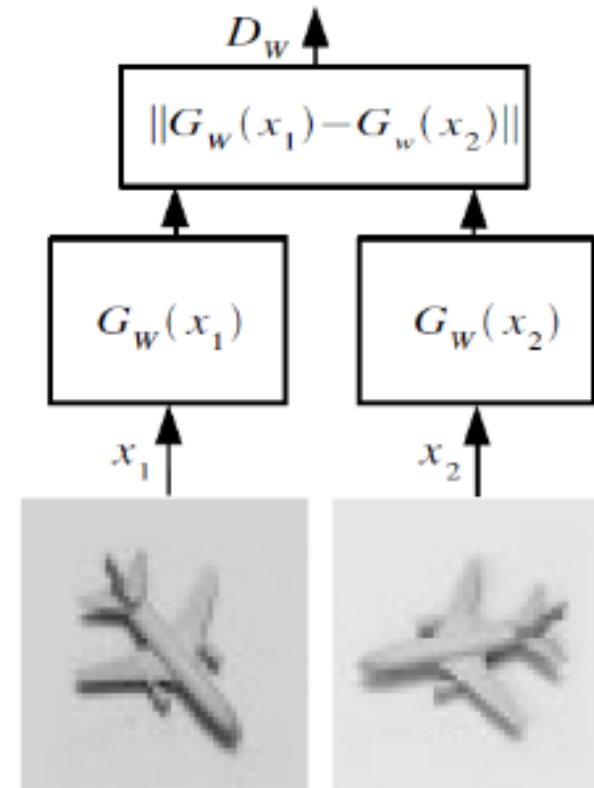
Siamese: Loss

Make this small



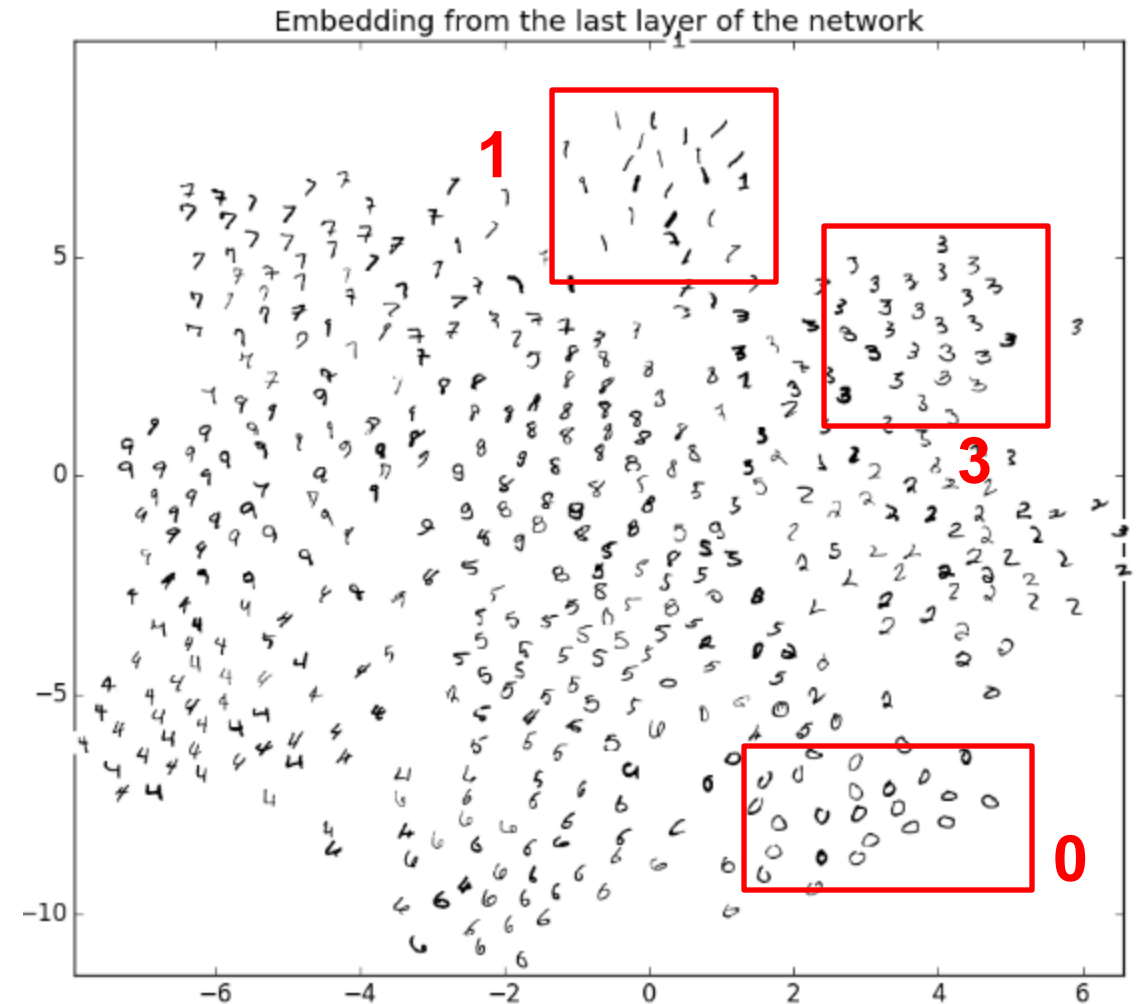
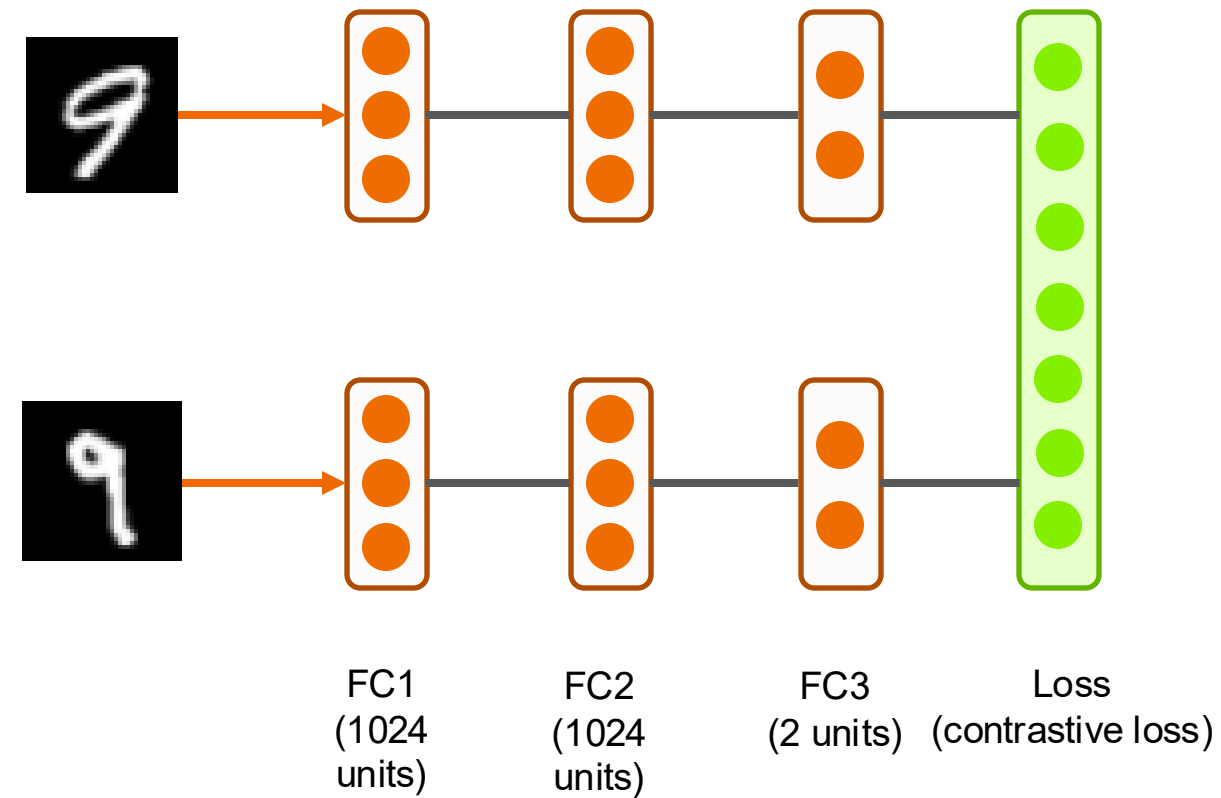
Similar images (neighbors
in the neighborhood graph)

Make this large

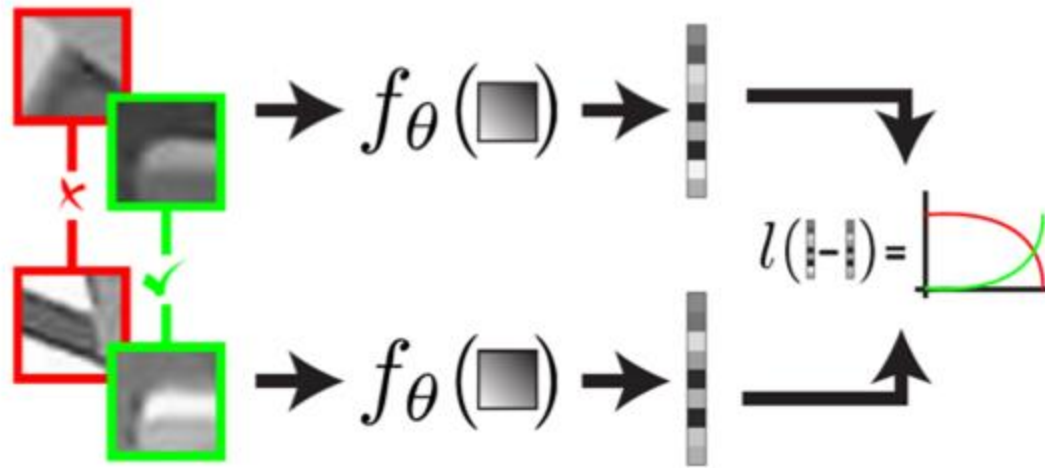


Dissimilar images
(non-neighbors in the
neighborhood graph)

Example: MNIST

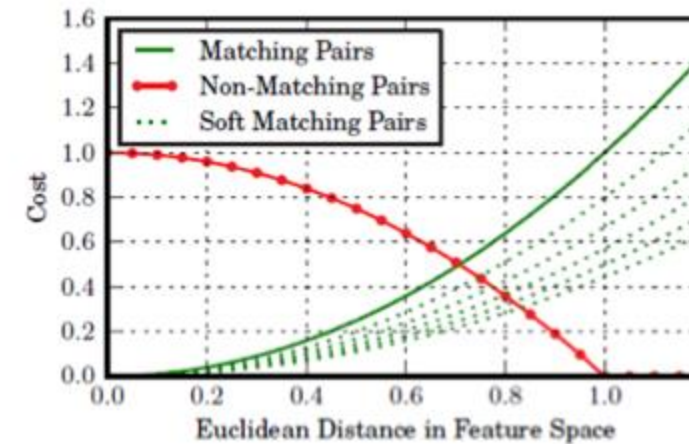


Application: Learning to Match Siamese Network

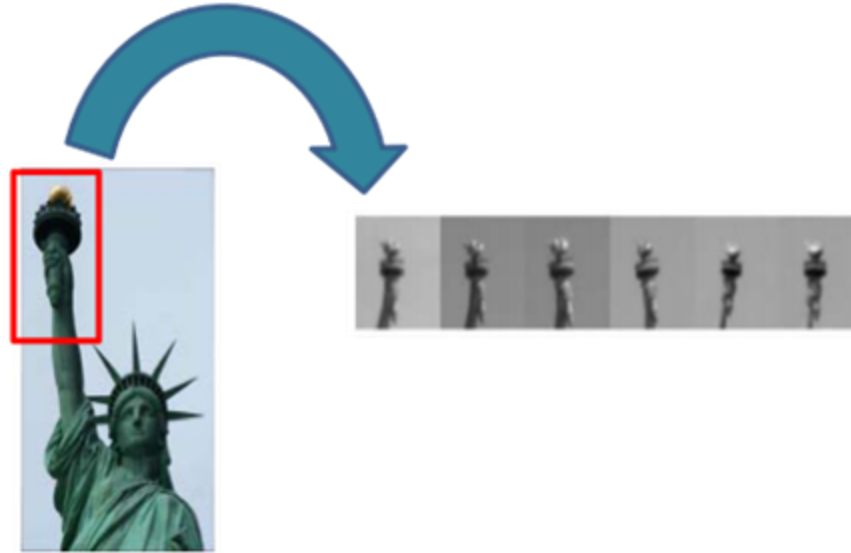


Using the contrastive cost function

$$l_{\theta}(y_i, y_j) = \begin{cases} s_{ij} d_{ij}^2, & \text{if matching} \\ \max(1.0 - d_{ij}^2, 0), & \text{if non-matching} \end{cases}$$



Matching across views



Learn a discriminative representation of patches from different views of 3D points

Smo-Serra, E, Trulls, E, Ferraz, L, Kokkinos, I., Fua, P. and Moreno-Noguer, F., 2015. Discriminative learning of deep convolutional feature point descriptors. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 118-126).

Application

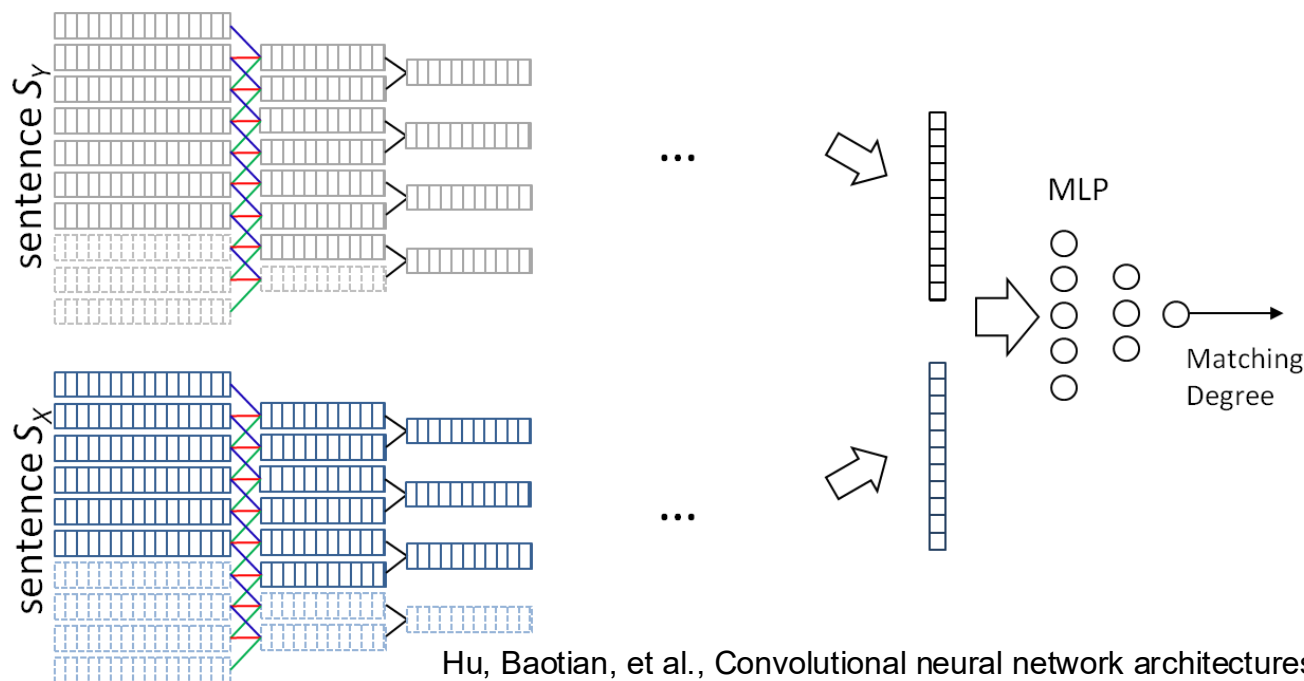
- **Application:** Sentence completion, response to tweet, paraphrase identification

Example:

word2vec



- x : Damn, I have to work overtime this weekend!
- Y⁺: Try to have some rest buddy.
- y: It is hard to find a job, better start polishing your resume.



Triplet Loss

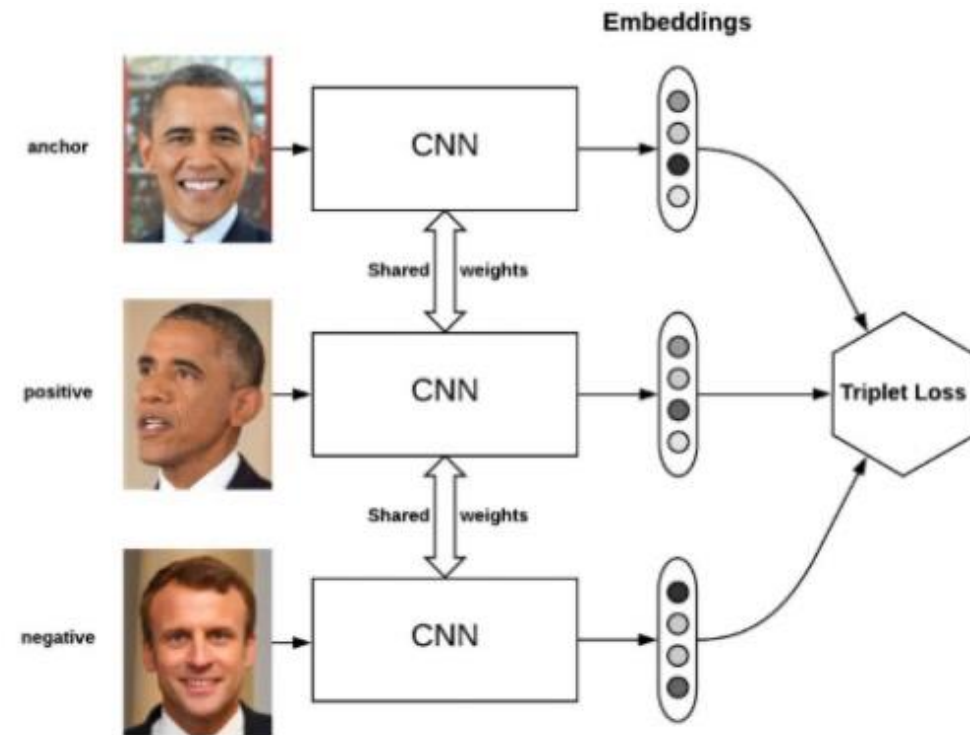
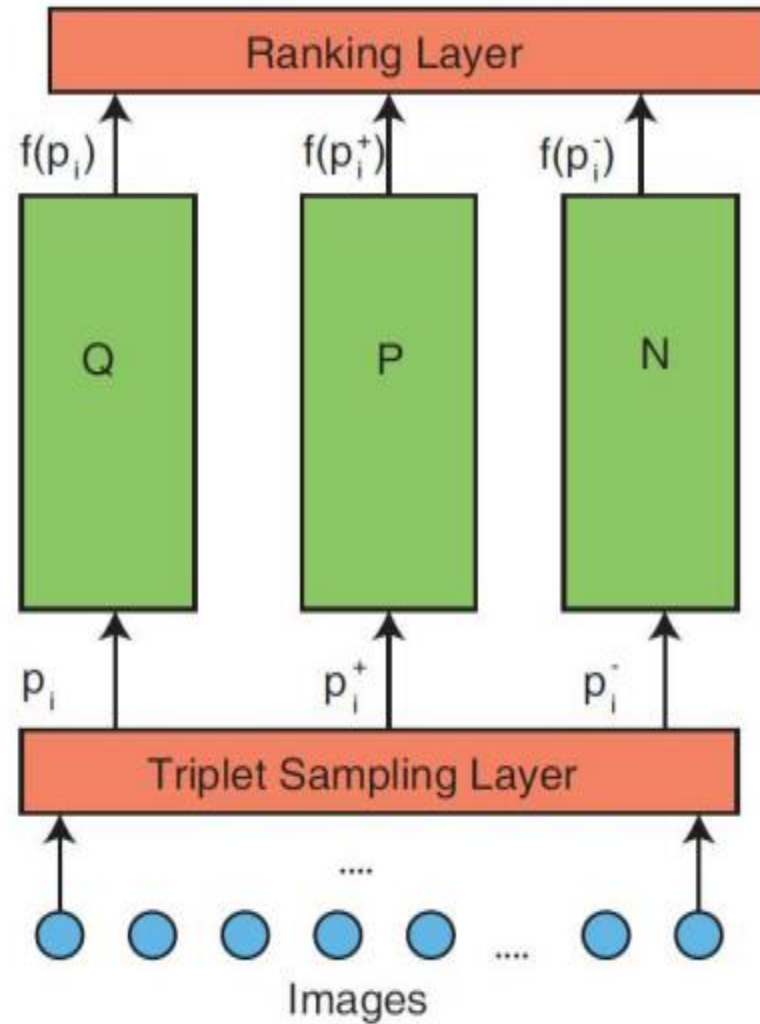
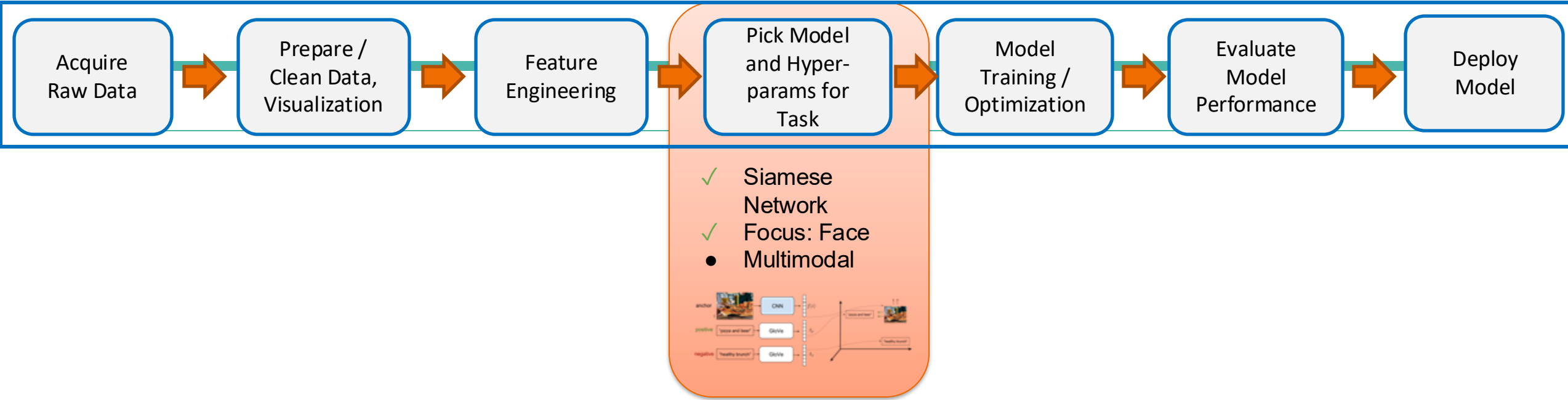


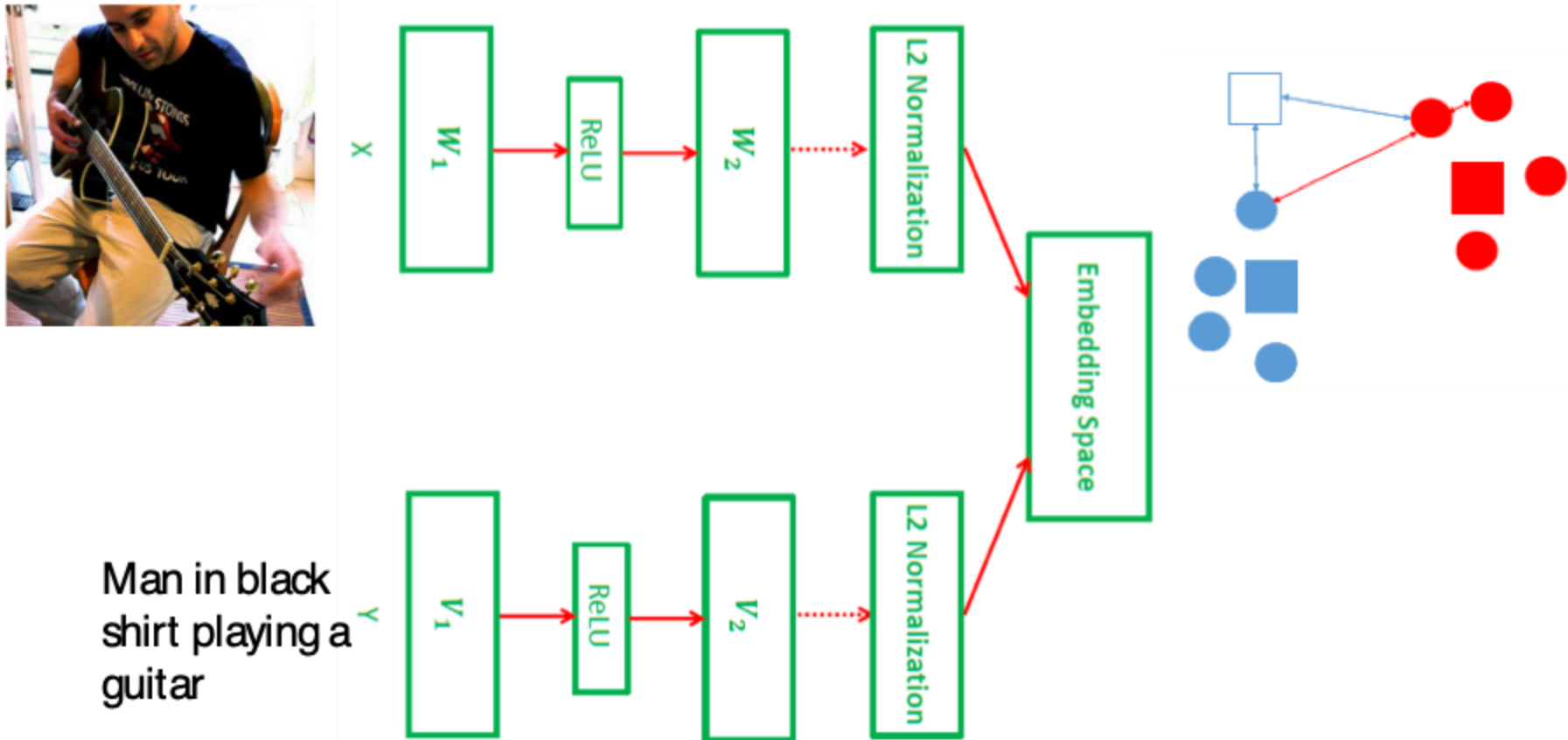
Image Retrieval (Ranking)

Query					
Positive					
Negative					



Extension: Multimodal

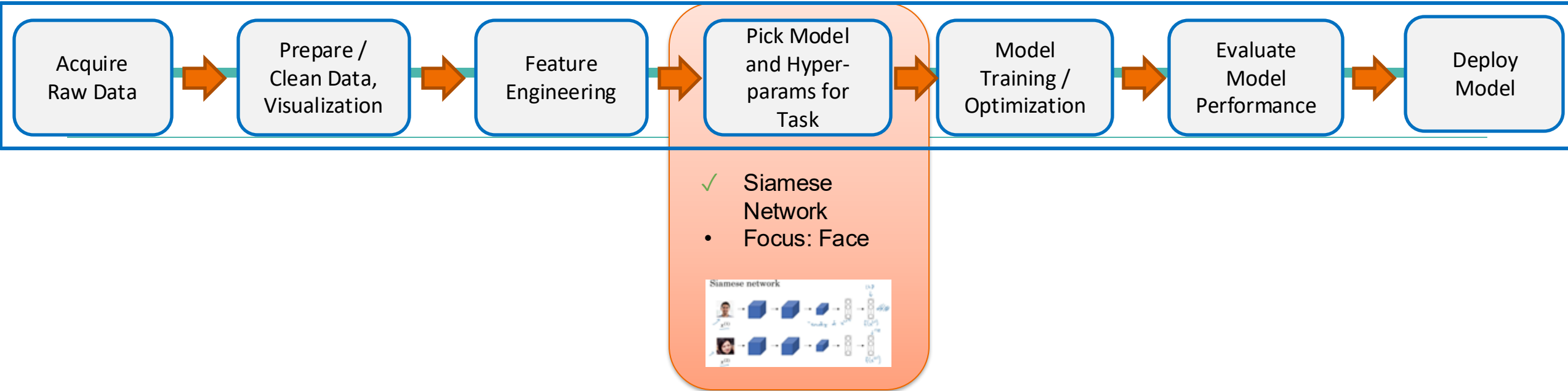
Cross modal matching



Wang, L., Li, Y. and Lazebnik, S., 2016. Learning deep structure-preserving image-text embeddings. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 5005-5013).

Summary

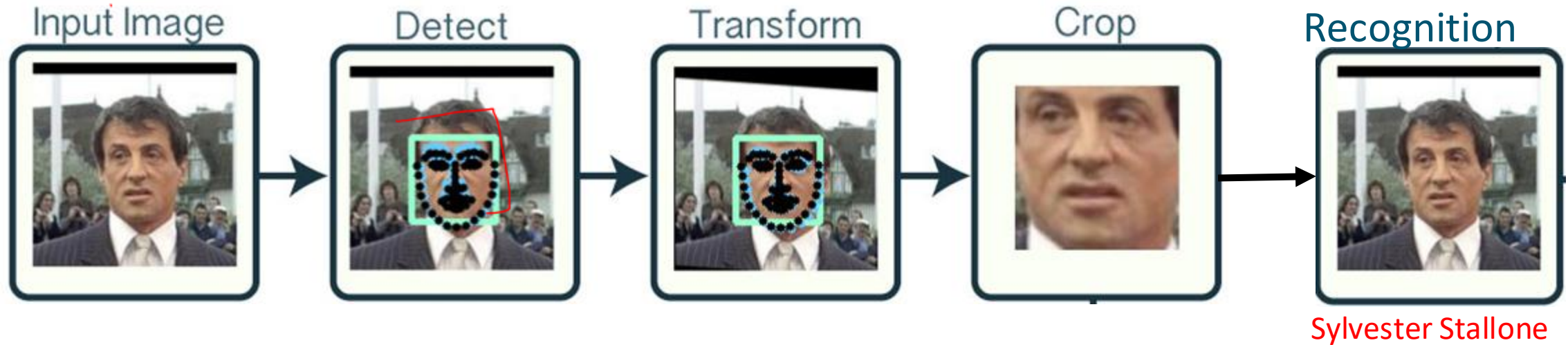
- Siamese and Triplet networks/losses are popular for solving fine grain classification (e.g. Face), capturing subjective needs (eg. Invariant to rotation), rankings etc.
- Available in many frameworks.
- Number of examples increase (e.g. nC_2 , nC_3). Finding right pairs/triplets is also important.



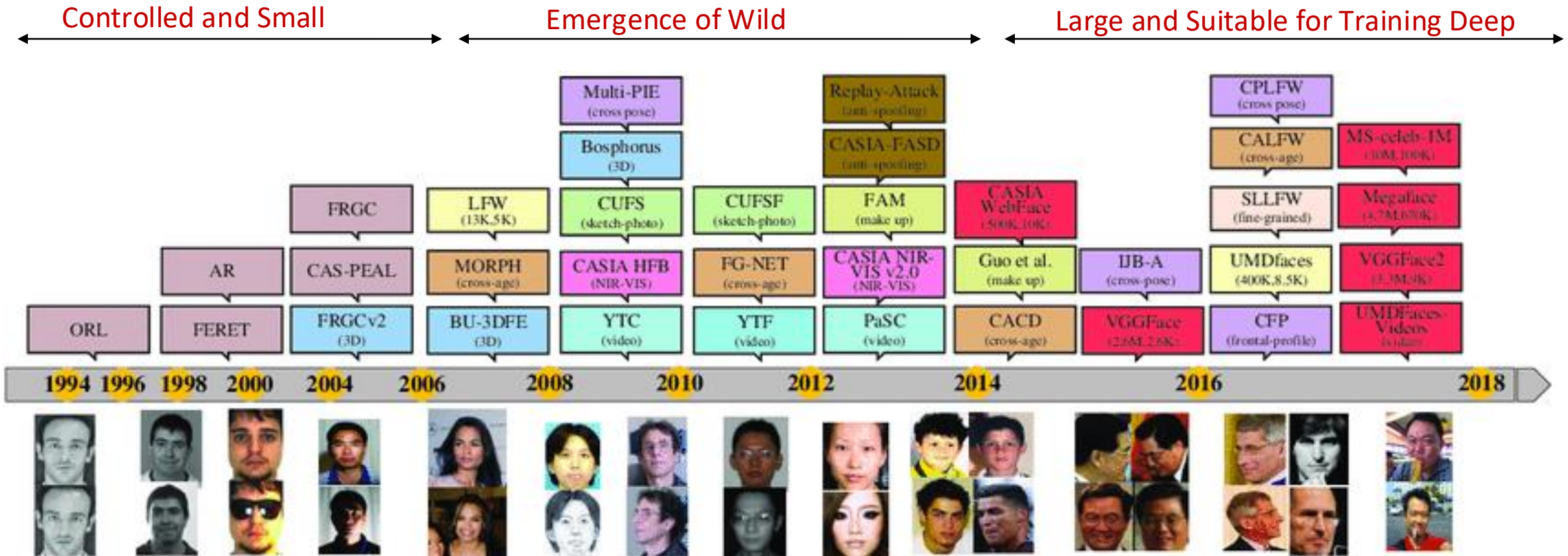
Example: Face

Focus on a specific area to appreciate the developments

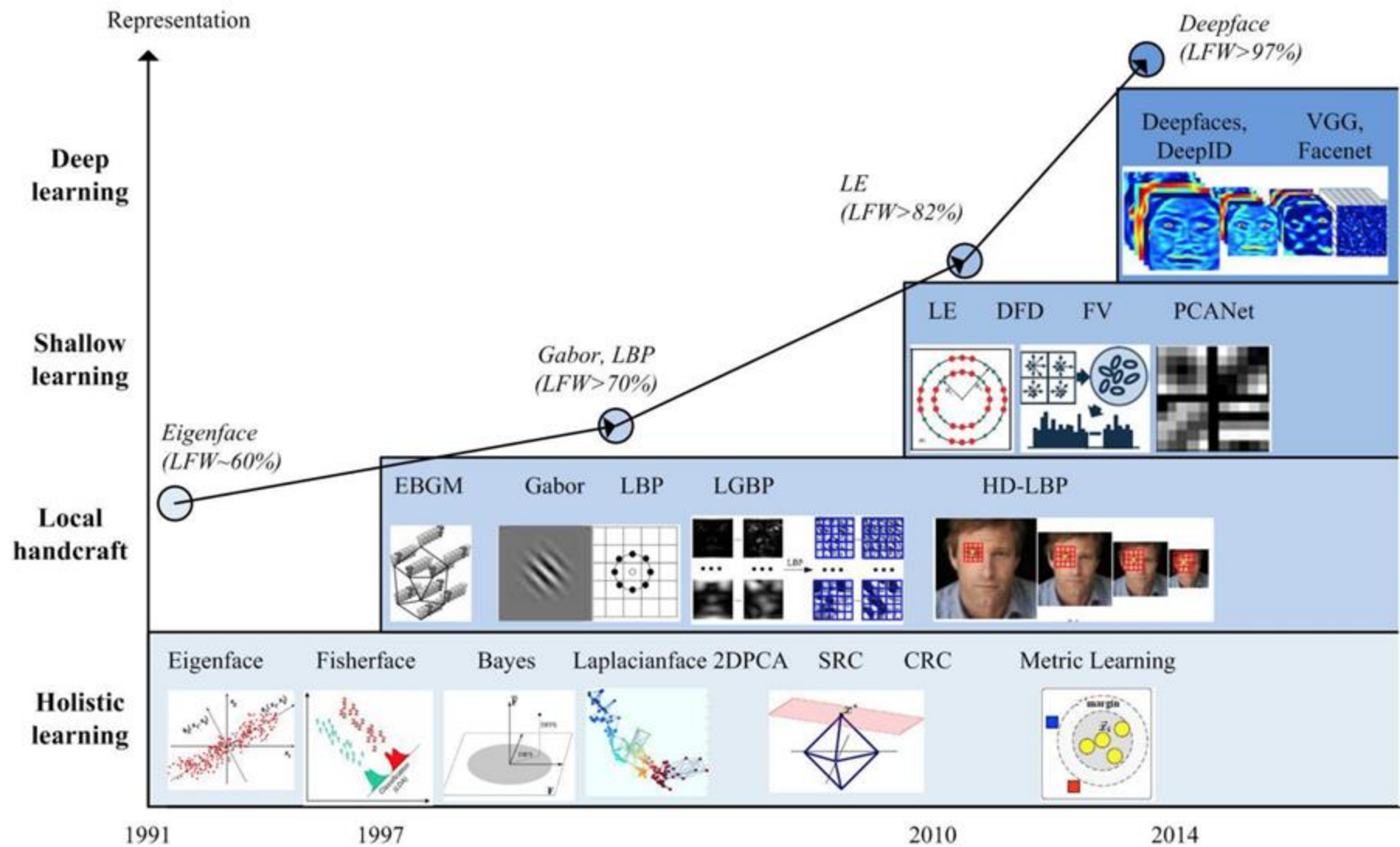
Face processing pipeline



Evolution of FR Datasets



Performance on LFW

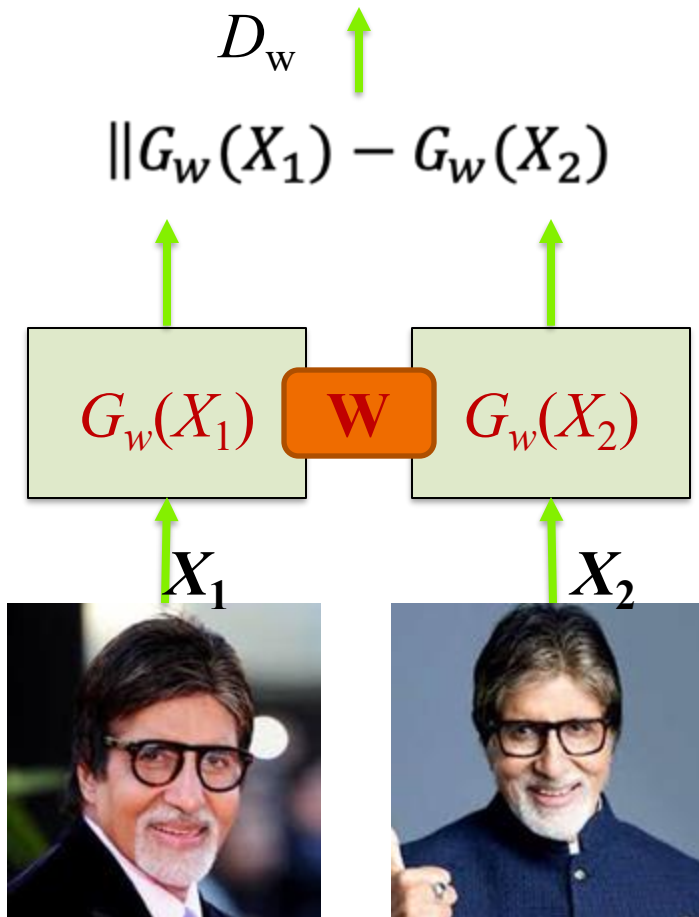


Systematic Evolution of Performance (LFW)

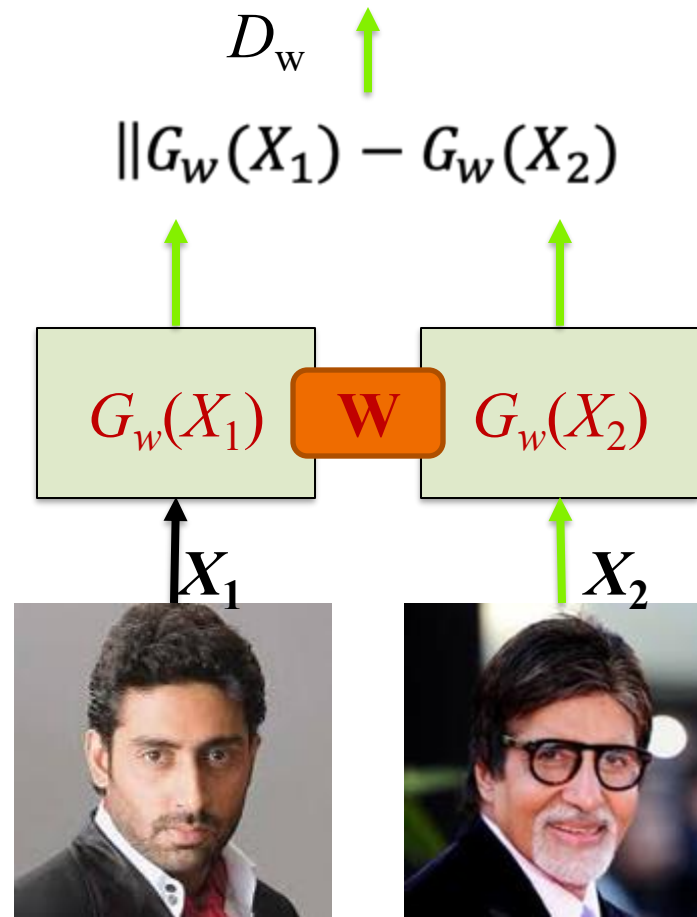
Method	Public. Time	Loss	Architecture	Number of Networks	Training Set	Accuracy $\pm$ Std(%)
DeepFace [20]	2014	softmax	Alexnet	3	Facebook (4.4M,4K)	97.35 $\pm$ 0.25
DeepID2 [21]	2014	contrastive loss	Alexnet	25	CelebFaces+ (0.2M,10K)	99.15 $\pm$ 0.13
DeepID3 [36]	2015	contrastive loss	VGGNet-10	50	CelebFaces+ (0.2M,10K)	99.53 $\pm$ 0.10
FaceNet [38]	2015	triplet loss	GoogleNet-24	1	Google (500M,10M)	99.63 $\pm$ 0.09
Baidu [58]	2015	triplet loss	CNN-9	10	Baidu (1.2M,18K)	99.77
VGGface [37]	2015	triplet loss	VGGNet-16	1	VGGface (2.6M,2.6K)	98.95
light-CNN [85]	2015	softmax	light CNN	1	MS-Celeb-1M (8.4M,100K)	98.8
Center Loss [101]	2016	center loss	Lenet+-7	1	CASIA-WebFace, CACD2000, Celebrity+ (0.7M,17K)	99.28
L-softmax [104]	2016	L-softmax	VGGNet-18	1	CASIA-WebFace (0.49M,10K)	98.71
Range Loss [82]	2016	range loss	VGGNet-16	1	MS-Celeb-1M, CASIA-WebFace (5M,100K)	99.52
L2-softmax [109]	2017	L2-softmax	ResNet-101	1	MS-Celeb-1M (3.7M,58K)	99.78
Normface [110]	2017	contrastive loss	ResNet-28	1	CASIA-WebFace (0.49M,10K)	99.19
CoCo loss [112]	2017	CoCo loss	-	1	MS-Celeb-1M (3M,80K)	99.86
vMF loss [115]	2017	vMF loss	ResNet-27	1	MS-Celeb-1M (4.6M,60K)	99.58
Marginal Loss [116]	2017	marginal loss	ResNet-27	1	MS-Celeb-1M (4M,80K)	99.48
SphereFace [84]	2017	A-softmax	ResNet-64	1	CASIA-WebFace (0.49M,10K)	99.42
CCL [113]	2018	center invariant loss	ResNet-27	1	CASIA-WebFace (0.49M,10K)	99.12
AMS loss [105]	2018	AMS loss	ResNet-20	1	CASIA-WebFace (0.49M,10K)	99.12
Cosface [107]	2018	cosface	ResNet-64	1	CASIA-WebFace (0.49M,10K)	99.33
Arcface [106]	2018	arcface	ResNet-100	1	MS-Celeb-1M (3.8M,85K)	99.83
Ring loss [117]	2018	Ring loss	ResNet-64	1	MS-Celeb-1M (3.5M,31K)	99.50

Siamese Loss

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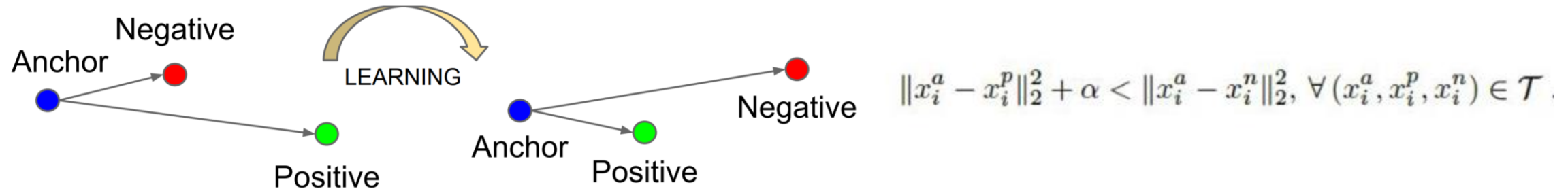


Make this larger



- Only pair-wise Labels
- Similarity Metric:
 $D_w(X_1, X_2)$
- Have shared weights
- Training in batches

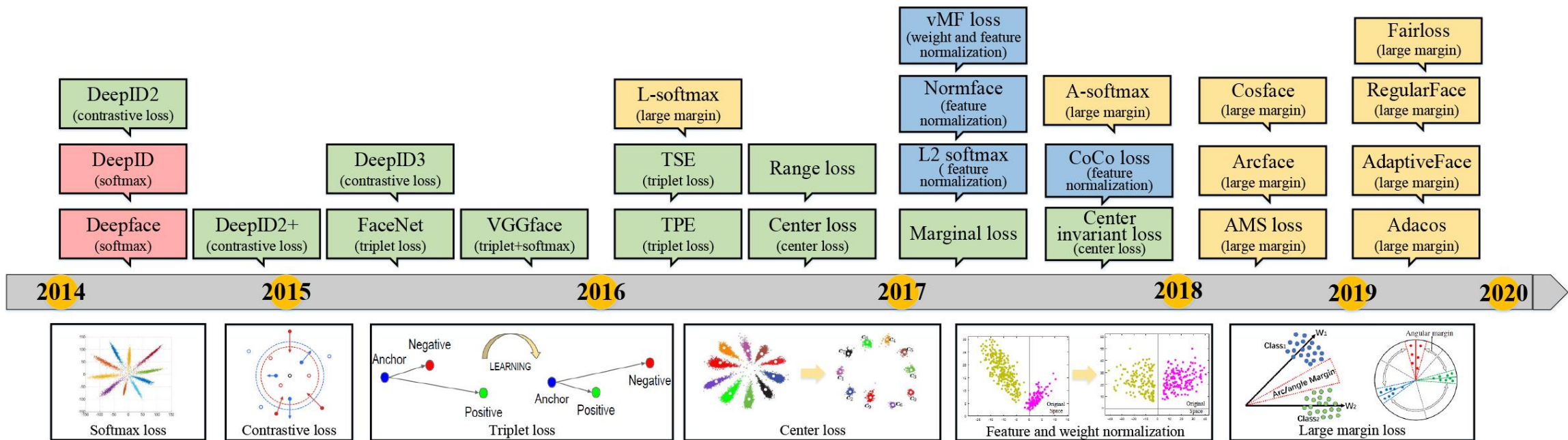
Triplet loss function



where f is the embedding

$$\sum_i^N \left[\|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha \right]_+$$

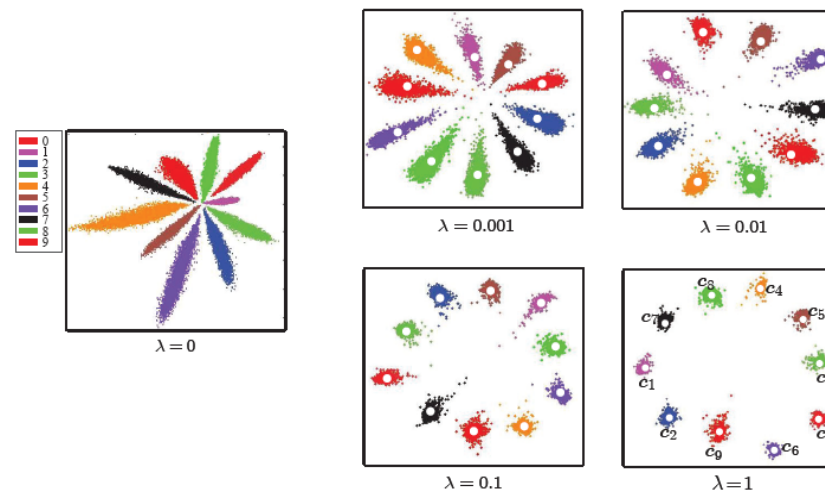
Summary: Evolution of Loss Functions



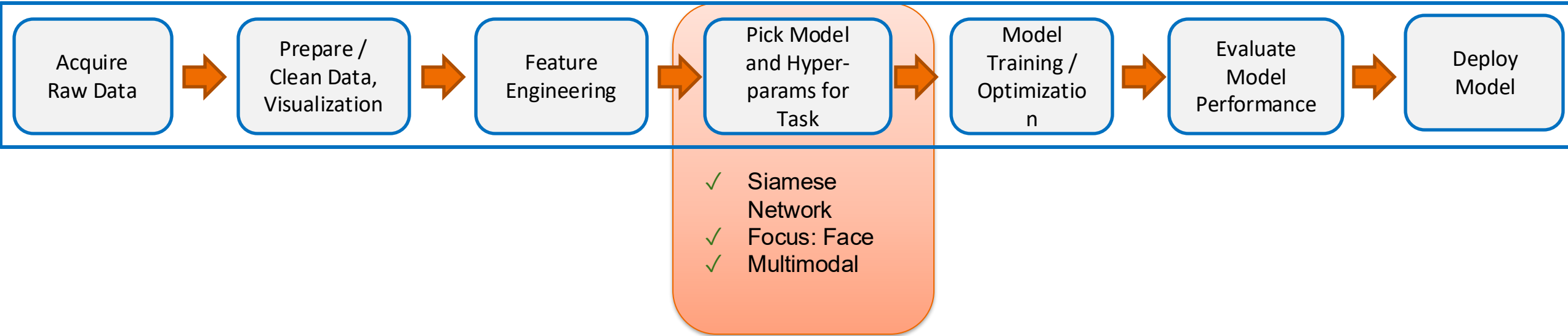
Center Loss

$$L = L_s + \mu L_c = -\sum_{i=1}^m \log \frac{e^{W_{y_i}^T x_i + b_{y_i}}}{\sum_{j=1}^n e^{W_j^T x_i + b_j}} + \frac{\lambda}{2} \sum_{i=1}^m ||x_i - c_{y_j}||_2^2$$

Two Terms (1) For Interclass separability (softmax) and (2) Intraclass compactness and Lambda that balance between them



Summary



Thanks!!

Questions?